Mediant Trees for Rational Approximation and Sturmian Word Generation

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Abstract

Given an oracle for a real number θ of sufficient precision, we present four contributions. First, an algorithm that constructs a *mediant tree* of increasingly precise rational approximations to θ: starting from a Farey-adjacent bounded pair bracketing θ, each iteration applies a weighted mediant step with an optimal integer weight w^* given by a closed-form formula (Theorem 8.1). The resulting weights naturally track the partial quotients of θ, allowing the algorithm to inherit superlinear denominator growth without a separate proof (Remark 10.2). Second, a Sturmian transduction that converts the mediant tree into a *mediant word* — a string over $\{\mathsf{O}, \mathsf{E}\}$ encoding the full approximation path — and establishes that this word is identical to the Sturmian word with slope $1/(1+\theta)$ at all interior positions, with the terminal symbol of each finite prefix fully characterised by the sign of Δ (Theorem 12.6). Third, we prove two structural invariants of mediant words: every non-leaf word begins with EE (Theorem 12.4), and the interior slice $\pi(r)[1{:}L{-}1]$ of any mediant word of length $L \geq 2$ is a palindrome (Theorem 12.5). Fourth, we show how to extend both algorithms to real numbers $\theta \in (0,1)$ and $\theta < 0$ via simple algebraic reductions to the $\theta > 1$ case.

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1 Introduction

The approximation of irrational numbers by rationals is a classical subject in number theory. The Stern–Brocot tree and Farey sequences provide a systematic account of best rational approximants, and the continued-fraction expansion of a real number yields its convergents — the fractions closest to the target with bounded denominator. Sturmian words, introduced by Morse and Hedlund [1], give a combinatorial counterpart: the infinite binary sequence $s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$ with slope $\alpha \in (0, 1)$ is the unique balanced aperiodic sequence with O-density α, and its finite prefixes are the *central words* studied by de Luca [2].

The present paper makes three contributions to this classical picture.

1. **Optimal weight formula.** We introduce the *bounded-pair* framework, in which two Farey-adjacent rational estimates bracket θ from above and below. At each step a weighted mediant is applied; Theorem 8.1 gives a closed-form expression for the optimal integer weight w, enabling efficient iteration with no redundant steps.

2. **Mediant word representation.** The mediant tree can be encoded as a string over $\{\mathsf{O}, \mathsf{E}\}$, the *mediant word* (Section 9). This representation connects rational approximation to combinatorics on words.

3. **Sturmian identity.** The mediant word produced by the mediant tree is identical to the Sturmian word with slope $1/(1+\theta)$ (Theorem 12.6), agreeing at every position except possibly the last symbol of each finite prefix. Consequently the mediant tree is an $O(\log N)$ compressed transducer for the Sturmian word: it encodes a prefix of length N using only $O(\log N)$ real-arithmetic operations, after which the symbols are enumerated by a purely integer recursive traversal.

The algorithm requires an oracle for θ — a means of evaluating $\lceil o\theta \rceil$ to sufficient precision at each step — but is otherwise entirely general. Given such an oracle, the mediant tree can represent *any* real number as a mediant word over $\{\mathsf{O}, \mathsf{E}\}$, making it a universal encoder from reals to binary words.

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Organisation. Part I (Sections 2–5) collects background on the Stern–Brocot tree, Farey sequences, Sturmian words, and de Luca central words. Part II develops the algorithm and its theory: Section 6 introduces the core definitions; Section 7 develops the mediant construction and straddling inequality; Section 8 derives the optimal weight formula; Section 9 introduces mediant words; Section 9.4 describes the mediant tree; Section 10 analyses the rate of convergence; Section 11 presents examples and computational results; Section 12 establishes the connection to Sturmian words; Section 13 extends the construction to all real $\theta \notin \{0, 1\}$; Section 14 concludes with open questions. Appendix A contains the proof of Theorem 8.1.

Part I
Background

2 The Stern–Brocot Tree and Mediants

The *Stern–Brocot tree* is a complete binary search tree containing every positive rational exactly once in lowest terms (see, e.g., Graham, Knuth, and Patashnik [3]). It is constructed by repeated *mediant* insertion: starting from the pair $(0/1, 1/0)$, the mediant of two adjacent fractions a/b and c/d is $(a+c)/(b+d)$, and inserting it between them yields two new adjacent pairs. The Stern–Brocot tree encodes continued-fraction expansions as paths from the root: a partial quotient of k corresponds to k steps in the same direction before the first reversal.

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Definition 2.1 (Farey adjacency)**.** Two fractions a/b and c/d (in lowest terms, $a/b < c/d$) are *Farey adjacent* if

$$bc - ad = 1.$$

Equivalently, they are adjacent in some Farey sequence F_n, or they lie on adjacent branches of the Stern–Brocot tree. Farey adjacency is equivalent to the condition that no rational with denominator at most $\max(b, d)$ lies strictly between a/b and c/d.

The mediant $(a + c)/(b + d)$ of two Farey-adjacent fractions is itself Farey adjacent to each of them. This adjacency-preservation is the key structural property exploited by our algorithm.

3 Farey Sequences

The *Farey sequence* F_n is the increasing sequence of fractions in $[0, 1]$ with denominator at most n, arranged in order. Its fundamental property is that consecutive terms $a/b < c/d$ in F_n satisfy $bc - ad = 1$ (i.e. they are Farey adjacent), and their mediant $(a + c)/(b + d)$ is the fraction with smallest denominator strictly between them.

For our purposes the key fact is the following: if a/b and c/d are Farey adjacent and $\theta \in (a/b, c/d)$, then every rational approximant to θ with denominator at most $b + d$ is one of a/b, c/d, or their mediant $(a + c)/(b + d)$. This is why maintaining Farey adjacency throughout the algorithm ensures that no improvement step is redundant.

4 Sturmian Sequences

Sturmian words were introduced by Morse and Hedlund [1] as the simplest aperiodic symbolic sequences. Let $\alpha \in (0,1)$ be irrational. The *lower Sturmian word* with slope α is defined by

$$s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor, \qquad n \geq 0.$$

Encoding $s_n = 1$ as O and $s_n = 0$ as E, this is a binary sequence with O-density exactly α.

Definition 4.1 (Sturmian word)**.** The *Sturmian word* $\mathbf{s}(\alpha)$ with slope $\alpha \in (0,1)$ is the infinite word over $\{\mathsf{O}, \mathsf{E}\}$ defined by $s_n = \mathsf{O}$ if $\lfloor (n+1)\alpha \rfloor > \lfloor n\alpha \rfloor$, and $s_n = \mathsf{E}$ otherwise.

Sturmian words are *balanced*: no two factors of the same length differ in their O-count by more than one. They are the unique balanced aperiodic sequences [1].

The two-word structure is fundamental: a Sturmian word with slope $\alpha \in (0, 1/2)$ consists of exactly two distinct *primitive words* (unbreakable building blocks), of lengths $\lfloor 1/\alpha \rfloor$ and $\lceil 1/\alpha \rceil$, concatenated in the Sturmian pattern. Concretely, with $\alpha = 1/(1+\theta)$ for $\theta > 1$, the two primitive words are $\mathsf{E}^{\lfloor \theta \rfloor}\mathsf{O}$ and $\mathsf{E}^{\lceil \theta \rceil}\mathsf{O}$ — exactly the leaf words of the mediant tree (Section 9.4).

Since $\alpha < 1/2$ for $\theta > 1$, the sequence avoids OO: two consecutive O's would require $\lfloor (n+2)\alpha \rfloor - \lfloor n\alpha \rfloor = 2$, but $2\alpha < 1$ makes this impossible. Conversely, when $\alpha > 1/2$ (i.e. $\theta < 1$), consecutive O's are unavoidable.

Sturmian morphisms. Sturmian words are invariant under the two *Sturmian morphisms* $\sigma_E \colon \mathsf{E} \mapsto \mathsf{E},\ \mathsf{O} \mapsto \mathsf{EO}$ and $\sigma_O \colon \mathsf{E} \mapsto \mathsf{OE},\ \mathsf{O} \mapsto \mathsf{O}$.

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5 Christoffel Words

De Luca [2] identified a distinguished class of *central words* as the palindromic cores of Sturmian prefixes.

Definition 5.1 (Central word)**.** A finite word w over $\{\mathsf{O}, \mathsf{E}\}$ is a *central word* if both $w\mathsf{EO}$ and $w\mathsf{OE}$ are Sturmian words. Every central word is a palindrome.

Definition 5.2 (Christoffel words)**.** For coprime positive integers e and o, the *lower Christoffel word* $\mathrm{Ch}^-(e,o)$ is the unique primitive word of length $L = e + o$ over $\{\mathsf{E}, \mathsf{O}\}$ containing exactly e copies of E and o copies of O that ends in O, with O-positions given by

$$\mathrm{Ch}^-(e,o)[i] = \mathsf{O} \iff \lfloor (i+1)o/L \rfloor > \lfloor i \cdot o/L \rfloor.$$

The *upper Christoffel word* $\mathrm{Ch}^+(e,o)$ is the reverse of $\mathrm{Ch}^-(e,o)$: it ends in E instead of O. Both words share the same central palindrome U as their interior: $\mathrm{Ch}^-(e,o) = \mathsf{E} \cdot U \cdot \mathsf{O}$ and $\mathrm{Ch}^+(e,o) = \mathsf{O} \cdot U \cdot \mathsf{E}$ (Berstel [6], Chapter 1).

$\mathrm{Ch}^-(e,o)$ is the lower Sturmian word of slope o/L truncated to length L and normalised to end in O; $\mathrm{Ch}^+(e,o)$ is the upper Sturmian word of the same slope, ending in E (Berstel [6], Chapter 1).

Each Christoffel word is characterised purely by its fraction o/L: given $\gcd(e, o) = 1$, the pair (e, o) is uniquely determined by $o/(e + o)$.

Theorem 5.3 (Christoffel concatenation (Berstel [6], Theorem 1.5.1))**.** *Let* $\mathrm{Ch}^-(e_0, o_0)$ *and* $\mathrm{Ch}^-(e_1, o_1)$ *be two lower Christoffel words with* $o_0/L_0 < o_1/L_1$ *(where* $L_k = e_k + o_k$*) satisfying the Farey adjacency condition* $o_1 L_0 - o_0 L_1 = 1$*. Then for any integer* $w \geq 1$*:*

$$\begin{aligned} \mathrm{Ch}^-(e_0, o_0)^w \cdot \mathrm{Ch}^-(e_1, o_1) &= \mathrm{Ch}^-(w\, e_0 + e_1,\ w\, o_0 + o_1), \\ \mathrm{Ch}^-(e_0, o_0) \cdot \mathrm{Ch}^-(e_1, o_1)^w &= \mathrm{Ch}^-(e_0 + w\, e_1,\ o_0 + w\, o_1). \end{aligned}$$

In the first identity the smaller-slope word is repeated; in the second the larger-slope word is repeated.

Part II
The Algorithm

6 Definitions and Notation

Throughout Sections 6–12, let $\theta > 1$ be a fixed real number. Section 13 handles all other real numbers by reduction to this case. We write $\mathbb{N}$ for the positive integers and $\lfloor \cdot \rfloor$, $\lceil \cdot \rceil$ for floor and ceiling.

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Remark 6.1 (Rational vs. irrational θ). Most of the combinatorial claims in these sections — the Christoffel word structure (Lemma 9.3), the palindrome invariant (Proposition 12.5), and the EE-prefix theorem (Proposition 12.4) — hold for all real $\theta > 1$, rational or not. The Sturmian identity (Theorem 12.6) and the last-letter theorem (Theorem 12.2) require θ to be irrational, since the Sturmian word is only well-defined for irrational slope $\alpha = 1/(1+\theta)$. The uniqueness of w^* (Theorem 8.1) also requires irrationality: for rational θ, $-A/B$ can be an integer, but in that case $K = 0$ and the algorithm terminates before w^* is needed. Near rational θ the algorithm terminates in finitely many steps (at the step where $K = 0$, i.e. where r_{i+1} is an exact representation of θ); all claims hold for every iterate up to termination.

Definition 6.2 (Ratio)**.** A *ratio* is a 2-tuple $r = (e, o) \in \mathbb{N}^2$, representing the rational number e/o. A ratio $r = (e, o)$ is a *rational estimate* of θ when used in that context.

Definition 6.3 (Mediant)**.** Given two ratios $r = (e, o)$ and $r' = (e', o')$ and a non-negative integer $w \geq 0$, the *mediant* of r and r' with weight w is the 3-tuple

$$\mathrm{Mediant}(r,\, r',\, w) = (r,\, r',\, w),$$

treated as a ratio by the composition rule

$$\mathrm{Mediant}(r,\, r',\, w) = (e + w\, e',\ o + w\, o').$$

The degenerate case $w = 0$ gives $\mathrm{Mediant}(r, r', 0) = r$. The Farey determinant of the pair (r, r') is

$$\Delta(r, r') = e'o - eo'.$$

Definition 6.4 (Good estimate)**.** An estimate e/o is a *good estimate* of θ if

$$e \in \{\lfloor o \cdot \theta \rfloor,\ \lceil o \cdot \theta \rceil\},$$

i.e. e/o is one of the two nearest rationals to θ with denominator o. A good estimate with $e = \lceil o\theta \rceil$ satisfies $e/o \geq \theta$ and is an *upper bound*; one with $e = \lfloor o\theta \rfloor$ satisfies $e/o \leq \theta$ and is a *lower bound*.

Definition 6.5 (Bounded pair)**.** Two good estimates e_l/o_l and e_u/o_u form a *bounded pair* if e_l/o_l is a lower bound, e_u/o_u is an upper bound,

$$\frac{e_l}{o_l} < \theta \leq \frac{e_u}{o_u},$$

and they satisfy the *Farey adjacency condition*

$$e_u \cdot o_l - e_l \cdot o_u = 1.$$

The Farey adjacency condition ensures that the pair lies on adjacent branches of the Stern–Brocot tree (Section 2) and that no rational with smaller denominator lies strictly between e_l/o_l and e_u/o_u.

Definition 6.6 (λ and ε)**.** For an estimate e/o of θ, define the *relative ratio*

$$\lambda = \frac{e/o}{\theta}$$

and the *relative error*

$$\varepsilon = \lambda - 1 = \frac{e/o}{\theta} - 1.$$

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An upper-bound good estimate satisfies $\lambda \geq 1$ (so $\varepsilon \geq 0$); a lower-bound good estimate satisfies $\lambda \leq 1$ (so $\varepsilon \leq 0$). In both cases $|\varepsilon|$ is small when o is large.

We will index estimates by $i, i+1, j, k$ and write λ_i, ε_i, etc. for the corresponding values. In a bounded pair we always orient so that $|\varepsilon_i| > |\varepsilon_{i+1}|$, i.e. r_{i+1} is the better estimate of the two.

Definition 6.7 (Δ and K)**.** Given a bounded pair $(r_i, r_{i+1}) = (e_i/o_i,\ e_{i+1}/o_{i+1})$ with $|\varepsilon_i| > |\varepsilon_{i+1}|$, define

$$\Delta = e_{i+1}o_i - e_i o_{i+1}$$

and

$$K = o_{i+1}(e_{i+1} - \theta \cdot o_{i+1}).$$

Geometrically, $K = o_{i+1}^2 \varepsilon_{i+1}$ is the signed absolute error of the better estimate r_{i+1} scaled by its denominator squared: it measures how far r_{i+1} sits from the target line $e = \theta o$ in units normalised to the lattice cell size.

The Farey adjacency condition in Definition 6.5 requires $e_u \cdot o_l - e_l \cdot o_u = 1$, which is precisely $|\Delta| = 1$ (the sign depends on which of r_i, r_{i+1} is the upper bound). One checks that $|K|/o_{i+1} = |e_{i+1} - \theta o_{i+1}| < 1$ (the good-estimate condition), and that $\operatorname{sgn}(K) = \operatorname{sgn}(\Delta)$ throughout the iteration.

Domain restriction. The definitions in this section assume $\theta > 1$. Section 13 shows how to reduce all other real numbers to this case.

Bounded pairs and the mediant tree. A bounded pair is a purely analytic object: any two good estimates that bracket θ and satisfy the Farey adjacency condition qualify, regardless of how they were found. The mediant construction (Section 7) provides a canonical way to *generate* bounded pairs, but the concept is wider. The invariants Δ and K belong to the pair itself; the Christoffel word structure (Section **??**) is a property of how the pair was reached via the mediant tree. In particular, $\Delta = \pm 1$ and $\operatorname{sgn}(K) = \operatorname{sgn}(\Delta)$ hold for any bounded pair, while the palindromic interior of $\pi(r)$ is a certificate that the estimate r was produced by the mediant algorithm.

7 The Mediant Construction

The classical *mediant* of two fractions e_i/o_i and e_{i+1}/o_{i+1} is $(e_i + e_{i+1})/(o_i + o_{i+1})$. We generalise this to a weighted family parametrised by a positive integer w.

Theorem 7.1 (Mediant convergence)**.** *Let (r_i, r_{i+1}) be a bounded pair for θ with $|\varepsilon_i| > |\varepsilon_{i+1}|$, and let $w \geq 1$ be a positive integer such that* $(\text{Mediant}(r_i, r_{i+1}, w-1),\ \text{Mediant}(r_i, r_{i+1}, w))$ *brackets θ. Write $r_{w-1} = \text{Mediant}(r_i, r_{i+1}, w-1)$ and $r_w = \text{Mediant}(r_i, r_{i+1}, w)$ for these two candidates. Then*

$$|\varepsilon_{w-1}| \leq |\varepsilon_i| \qquad \text{and} \qquad |\varepsilon_w| < |\varepsilon_{i+1}| < |\varepsilon_i|.$$

(When $w = 1$, $r_{w-1} = r_i$ exactly so $|\varepsilon_{w-1}| = |\varepsilon_i|$; otherwise $|\varepsilon_{w-1}| < |\varepsilon_i|$.) Moreover (r_{w-1}, r_w) is itself a bounded pair. The two mediants are assigned as r_{i+2} and r_{i+3}, where r_{i+3} is whichever has the smaller $|\varepsilon|$ (the better estimate), and r_{i+2} takes the other. Since r_w always satisfies $|\varepsilon_w| < |\varepsilon_{i+1}|$, we have $|\varepsilon_{i+3}| < |\varepsilon_{i+1}|$ regardless of which candidate is assigned to r_{i+3}.

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Proof. The denominators satisfy $o_{w-1} = o_i + (w-1)o_{i+1} < o_i + w\,o_{i+1} = o_w$, so r_{w-1} and r_w are distinct. The ratio λ_{w-1} is a weighted combination of λ_i and λ_{i+1}:

$$\lambda_{w-1} = \frac{o_i\,\lambda_i + (w-1)o_{i+1}\,\lambda_{i+1}}{o_i + (w-1)o_{i+1}}.$$

When $w = 1$ this gives $\lambda_{w-1} = \lambda_i$ exactly, so $|\varepsilon_{w-1}| = |\varepsilon_i|$. When $w \geq 2$, λ_{w-1} is a strict convex combination of λ_i and λ_{i+1}, lying strictly between them, so $|\lambda_{w-1} - 1| < |\lambda_i - 1|$, i.e. $|\varepsilon_{w-1}| < |\varepsilon_i|$. For r_w (weight $w \geq 1$), the combination is always strict, so $|\lambda_w - 1| < |\lambda_i - 1|$ and $|\lambda_w - 1| < |\lambda_{i+1} - 1|$, i.e. $|\varepsilon_w| < |\varepsilon_{i+1}| < |\varepsilon_i|$. The Farey adjacency condition for (r_{w-1}, r_w) follows by direct expansion:

$$\begin{aligned} e_w o_{w-1} - e_{w-1} o_w &= (e_i + w\,e_{i+1})(o_i + (w-1)o_{i+1}) - (e_i + (w-1)e_{i+1})(o_i + w\,o_{i+1}) \\ &= \big[(w-1) - w\big] e_i\,o_{i+1} + \big[w - (w-1)\big] e_{i+1}\,o_i \\ &= -e_i\,o_{i+1} + e_{i+1}\,o_i = \Delta, \end{aligned}$$

so $|e_w o_{w-1} - e_{w-1} o_w| = |\Delta| = 1$, and the Farey adjacency of (r_{w-1}, r_w) is inherited directly from that of (r_i, r_{i+1}). □

Corollary 7.2 (Straddling inequality)**.** *Under the conditions of Theorem 7.1,*

$$\varepsilon_{w-1}\,\varepsilon_w < 0 \qquad \textit{and} \qquad |\varepsilon_{w-1}||\varepsilon_w| < |\varepsilon_i|\,|\varepsilon_{i+1}|.$$

Proof. The first inequality holds because r_{w-1} and r_w straddle θ, so their errors have opposite signs. For the second: $|\varepsilon_{w-1}| \le |\varepsilon_i|$ and $|\varepsilon_w| < |\varepsilon_{i+1}|$ (Theorem 7.1), so $|\varepsilon_{w-1}||\varepsilon_w| \le |\varepsilon_i||\varepsilon_w| < |\varepsilon_i|\,|\varepsilon_{i+1}|$. □

8 The Optimal Weight Formula

A key question is: for which integer $w \ge 1$ do the two consecutive weighted mediants

$$r_j = \text{Mediant}(r_i, r_{i+1}, w-1), \qquad r_k = \text{Mediant}(r_i, r_{i+1}, w)$$

straddle θ, i.e. satisfy the bracketing condition $(\lambda_j - 1)(\lambda_k - 1) \le 0$? (Here r_j and r_k are local names for the two candidates at weights $w-1$ and w; they are used only within this section and Section A.)

Writing $A = e_i - \theta\, o_i$ and $B = e_{i+1} - \theta\, o_{i+1}$ for the scaled errors, the bracketing condition becomes $(A + (w-1)B)(A + wB) < 0$. Observe that $A + (w-1)B$ and $A + wB$ are consecutive evaluations of the linear function $f(w) = A + wB$ at $w-1$ and w: their product is negative iff f changes sign between $w-1$ and w, i.e. iff the zero $-A/B$ of f lies in the interval $(w-1, w)$, i.e. iff $w = \lfloor -A/B \rfloor + 1$. Since θ is irrational, $-A/B$ is never an integer, so exactly one such w exists. The unique integer satisfying this bracketing condition is given by the following closed-form expression.

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Theorem 8.1 (Optimal weight)**.** *Let (r_i, r_{i+1}) be a bounded pair bracketing θ, with*

$$A = e_i - \theta\, o_i, \qquad B = e_{i+1} - \theta\, o_{i+1},$$

and set $\Delta = e_{i+1}o_i - e_i o_{i+1}$, $K = o_{i+1}B$. Then the unique integer $w \ge 1$ for which $r_j = \text{Mediant}(r_i, r_{i+1}, w-1)$ and $r_k = \text{Mediant}(r_i, r_{i+1}, w)$ satisfy the bracketing condition $(\lambda_j - 1)(\lambda_k - 1) \le 0$ is

$$w^* = \left\lfloor \frac{-A}{B} \right\rfloor + 1 = \left\lfloor \frac{\Delta}{K} - \frac{o_i}{o_{i+1}} \right\rfloor + 1.$$

A derivation is given in Appendix A.

When $K = 0$, r_{i+1} is an exact rational representation of θ and no further improvement is possible; the algorithm terminates.

The continuous optimum and the discrete formula coincide

To see why w^* is the natural choice, consider treating w as a real variable and finding the stationary point of the error product $f(w-1)\,f(w)$, where $f(w) = A + wB$. Differentiating with respect to w and setting the derivative to zero:

$$\frac{d}{dw}\big[f(w-1)\,f(w)\big] = B\big[(A + wB) + (A + (w-1)B)\big] = 0.$$

Solving gives the unique real critical point

$$w_0 = \tfrac{1}{2} - \frac{A}{B} = c + \tfrac{1}{2},$$

where $c = -A/B > 0$. The product $f(w-1)\,f(w)$ has roots at $w = c$ (where $f(w) = 0$) and $w = c+1$ (where $f(w-1) = 0$), and $w_0 = c + \frac{1}{2}$ is their midpoint — the unique real maximum of $|f(w-1)\,f(w)|$ between the roots. The unique integer w for which $f(w-1)$ and $f(w)$ have opposite signs satisfies $c < w < c+1$, i.e. $w = \lfloor c \rfloor + 1 = w^*$. Since $w_0 = c + \frac{1}{2}$ also lies in $(c,\, c+1)$, and w^* is the unique integer in that interval, w^* is the nearest integer to w_0: both the continuous stationary point and the sign-change bracket single out the same w^*.

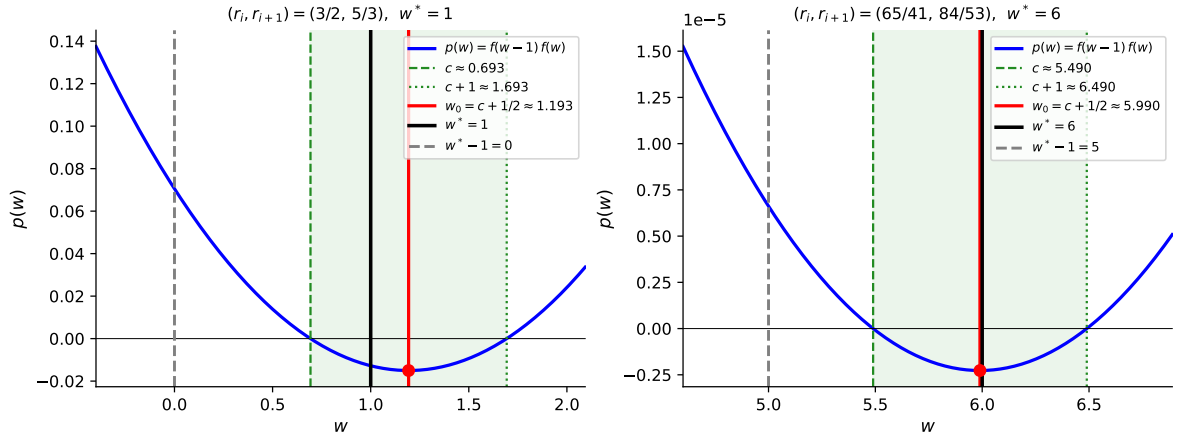


Figure 1: The product $p(w) = f(w-1)\,f(w)$ for two pairs from the mediant tree for $\theta = \log_2 3$, illustrating the optimal weight w^*. *Left:* pair (3/2, 5/3) with $c \approx 0.693$ and $w^* = 1$. *Right:* pair (65/41, 84/53) with $c \approx 5.490$ and $w^* = 6$. In each case the roots of p are at c (where $f(w) = 0$, green dashed) and $c+1$ (where $f(w-1) = 0$, green dotted); the stationary point $w_0 = c + \frac{1}{2}$ (red) is their midpoint; the unique integer in the shaded interval $(c,\, c+1)$ is w^* (black); and its companion $w^* - 1$ (grey dashed) lies just outside the interval on the left. Together $w^* - 1$ and w^* are the two weights whose mediants bracket θ.

Algorithm. The iteration proceeds as follows.

1. Start with an initial bounded pair (r_0, r_1) for θ.
2. Compute Δ and K.
3. If $K = 0$, terminate.
4. Compute $w^* = \lfloor \Delta/K - o_i/o_{i+1} \rfloor + 1$.
5. Form the new bounded pair $(\text{Mediant}(r_i, r_{i+1}, w^*-1),\ \text{Mediant}(r_i, r_{i+1}, w^*))$ and repeat from step 2.

Because o_{i+1} grows monotonically along the iteration, the denominator of K grows and the iteration converges (Section 10).

9 Mediant Words and the Mediant Tree

The mediant construction admits a compact combinatorial encoding. Every estimate produced by the algorithm is either a *leaf* or a *branch*, and the recursive structure of all estimates defines a binary tree whose nodes carry both rational values and combinatorial words over $\{\mathsf{O}, \mathsf{E}\}$.

9.1 Leaf words

Definition 9.1 (Leaf mediant word)**.** Let $\{\mathsf{O}, \mathsf{E}\}^*$ denote the set of finite strings over the two-letter alphabet $\{\mathsf{O}, \mathsf{E}\}$. A *leaf* estimate has denominator $o = 1$, i.e. it is a base estimate $e/1$ for some positive integer e. Its *mediant word* is

$$\pi(e/1) = \underbrace{\mathsf{E} \cdots \mathsf{E}}_{e} \mathsf{O},$$

a string of e copies of E followed by one O.

The two leaf estimates that initialise the algorithm are $r_0 = \lfloor \theta \rfloor / 1$ and $r_1 = \lceil \theta \rceil / 1$, with mediant words $\pi(r_0) = \mathsf{E}^{\lfloor \theta \rfloor}\mathsf{O}$ and $\pi(r_1) = \mathsf{E}^{\lceil \theta \rceil}\mathsf{O}$. Since θ is irrational, $\lceil \theta \rceil = \lfloor \theta \rfloor + 1$, so $\pi(r_0)$ and $\pi(r_1)$ differ by exactly one E. Each word $\mathsf{E}^k\mathsf{O}$ contains exactly one O, so it is not a power of a shorter word; both $\pi(r_0)$ and $\pi(r_1)$ are therefore primitive. Moreover, $\mathsf{E}^k\mathsf{O}$ is the lower Christoffel word $\mathrm{Ch}^-(k, 1)$ (Definition 5.2): its O appears at position k, matching the Christoffel formula. Thus both leaf words are lower Christoffel words of their respective fractions.

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9.2 Branch words and the weighted operators

Each branch step produces two new mediant words from the current pair $(u, v) = (\pi(r_i), \pi(r_{i+1}))$ by concatenation:

Definition 9.2 (Branch mediant word)**.** A *branch* estimate r_{i+2} or r_{i+3} has two *ordered parents* (r_i, r_{i+1}) and a positive integer *weight* w. Estimates are indexed so that r_{2n} and r_{2n+1} form the bounded pair at depth n, with r_{2n+1} always the better estimate (smaller $|\varepsilon|$). Write $u = \pi(r_i)$ and $v = \pi(r_{i+1})$ for the parent mediant words.

The two branch words at each depth share the same parents and differ only in the multiplicity of v: one estimate uses weight w and the other uses weight $w - 1$. Which is the better estimate depends on the errors: writing ε_j and ε_k for the errors of the two candidates (Definition 6.6),

$$w_{i+3} - w_{i+2} = \begin{cases} +1 & \text{if } |\varepsilon_k| < |\varepsilon_j|, \\ -1 & \text{otherwise.} \end{cases}$$

In either case $|\pi(r_{i+3})| - |\pi(r_{i+2})| = \pm|v|$.

The branch mediant words are:

- $\Delta > 0$ (r_{i+1} is the upper bound, v plays the role of E):

$$\pi(r_{i+2}) = v^{w-1} \cdot u,$$
$$\pi(r_{i+3}) = v^{w} \cdot u.$$

- $\Delta < 0$ (r_{i+1} is the lower bound, v plays the role of O):

$$\pi(r_{i+2}) = u \cdot v^{w-1},$$
$$\pi(r_{i+3}) = u \cdot v^{w}.$$

Which of the two words is the better estimate r_{i+3} is determined by comparing $|\varepsilon_j|$ and $|\varepsilon_k|$; the assignment is not fixed in advance.

Concrete example ($w = 2$, depth-0 leaves $\pi(r_0) = \mathtt{EO}$, $\pi(r_1) = \mathtt{EEO}$, $\Delta > 0$):

$$\Phi^2\big(\mathtt{EEO},\, \mathtt{EO}\big) = \big(\mathtt{EEO}^2\,\mathtt{EO},\ \mathtt{EEO}\big) = \big(\mathtt{EEOEEOEO},\ \mathtt{EEO}\big),$$

so $\pi(r_3) = \mathtt{EEOEEOEO}$ and $\pi(r_2) = \mathtt{EEOEO}$ ($= \Phi^1(\mathtt{EEO}, \mathtt{EO})$, the $w{-}1$ companion).

9.3 Mediant words are Christoffel words

A word is *primitive* if it is not a power x^k of a strictly shorter word x (with $k \geq 2$).

Lemma 9.3 (Every mediant word is a lower Christoffel word)**.** *For every estimate $r = (e, o)$ produced by the mediant tree, $\pi(r) = \mathrm{Ch}^-(e, o)$, the lower Christoffel word at slope $o/(e+o)$. In particular, $\pi(r)$ is never an upper Christoffel word, since by Theorem 12.1 every mediant word ends in O, whereas upper Christoffel words end in E.*

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Proof. By structural induction on the mediant tree.

Base case. By §9.1, $\pi(e/1) = \mathsf{E}^e\mathsf{O} = \mathrm{Ch}^-(e, 1)$.

Inductive step. Let (r_i, r_{i+1}) be a bounded pair with $\Delta = e_{i+1}o_i - e_i o_{i+1}$. By induction $\pi(r_i) = \mathrm{Ch}^-(e_i, o_i)$ and $\pi(r_{i+1}) = \mathrm{Ch}^-(e_{i+1}, o_{i+1})$, both lower Christoffel words (ending in O), with slopes o_i/L_i and o_{i+1}/L_{i+1} (where $L_k = e_k + o_k$) on opposite sides of $\alpha = 1/(1+\theta)$.

When $\Delta > 0$: r_{i+1} is the upper estimate so $e_{i+1}/o_{i+1} > \theta$, giving $o_{i+1}/L_{i+1} < \alpha < o_i/L_i$. The new word is $v^w u = \pi(r_{i+1})^w \cdot \pi(r_i)$: the smaller-slope word $\pi(r_{i+1})$ repeated w times, followed by $\pi(r_i)$ once. By Theorem 5.3 (with $e_0 = e_{i+1}$, $e_1 = e_i$) this equals $\mathrm{Ch}^-(w\,e_{i+1} + e_i,\ w\,o_{i+1} + o_i)$.

When $\Delta < 0$: r_{i+1} is the lower estimate so $e_{i+1}/o_{i+1} < \theta$, giving $o_i/L_i < \alpha < o_{i+1}/L_{i+1}$. Thus $\pi(r_i)$ has the smaller slope and $\pi(r_{i+1})$ has the larger slope. The new word is $uv^w = \pi(r_i) \cdot \pi(r_{i+1})^w$: the single smaller-slope word followed by the larger-slope word repeated w times. The applicable identity is the symmetric companion:

$$\mathrm{Ch}^-(e_0, o_0) \cdot \mathrm{Ch}^-(e_1, o_1)^w = \mathrm{Ch}^-(e_0 + w\,e_1,\ o_0 + w\,o_1) \quad (o_0/L_0 < o_1/L_1),$$

(Berstel [6], Theorem 1.5.1), giving $\mathrm{Ch}^-(e_i + w\,e_{i+1},\ o_i + w\,o_{i+1})$.

In both cases $\gcd(e_i + we_{i+1},\ o_i + wo_{i+1}) = 1$ by the Farey adjacency condition $|e_{i+1}o_i - e_i o_{i+1}| = 1$, confirming the result is a valid lower Christoffel word. □

The Christoffel identification is the key fact connecting the mediant algorithm to the combinatorics of Sturmian words. The recurrence $\pi(r_{i+3}) = v^w \cdot u$ (or $u \cdot v^w$) matches the standard sequence generation rule for Sturmian words (Widmer [5], Proposition 2.4.6), with weight w encoding the partial quotient at that depth.

9.4 The mediant tree

The mediant tree is the practical device for generating mediant words of arbitrary length and precision. Given a target θ and a desired precision, one runs the algorithm to depth n, obtaining a bounded pair (r_{2n}, r_{2n+1}) whose mediant words have length $e_{2n+1} + o_{2n+1}$. The mediant word $\pi(r_{2n+1})$ is then read off the tree by the recursive rule: an interior node Mediant(r_0, r_1, w) with $\Delta > 0$ yields $\pi(r_1)^w \cdot \pi(r_0)$, and with $\Delta < 0$ yields $\pi(r_0) \cdot \pi(r_1)^w$. Deeper trees produce longer prefixes of the Sturmian word with slope $1/(\theta+1)$ (Section 12).

Definition 9.4 (Mediant tree)**.** The *mediant tree* of an estimate r is defined recursively:

- If $r \in \{\lfloor\theta\rfloor, \lceil\theta\rceil\}$, then r is a *leaf node*.
- Otherwise r is an *interior node* Mediant$(r_0,\, r_1,\, w)$ with left subtree r_0, right subtree r_1, and integer weight $w \geq 0$, representing the rational

$$r = \frac{e_0 + w\, e_1}{o_0 + w\, o_1}.$$

The two leaves are the unique consecutive integers bracketing θ, forming the initial bounded pair $(r_0, r_1) = (\lfloor\theta\rfloor, \lceil\theta\rceil)$. Each bounded pair (r_{2n}, r_{2n+1}) in the algorithm's sequence consists of two mediant trees sharing the same leaves; as n increases the trees grow deeper, recording the full construction history of each estimate.

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Example 9.5 (Mediant words and trees for $\log_2 3$)**.** Starting from the initial bounded pair $(r_0, r_1) = (1/1,\, 2/1)$ with $\theta = \log_2 3 \approx 1.58496$, the algorithm produces the following sequence of bounded pairs. At each step n, weight w_{2n+1} is used to compute the two candidates $r_j = \text{Mediant}(r_{2n}, r_{2n+1}, w_{2n+1} - 1)$ and $r_k = \text{Mediant}(r_{2n}, r_{2n+1}, w_{2n+1})$; the better estimate (smaller $|\varepsilon|$) becomes r_{2n+3}.

n	r_{2n}	r_{2n+1}	$\varepsilon_{r_{2n}}$	$\varepsilon_{r_{2n+1}}$	w_{2n+1}	r_j	r_k
0	1/1	2/1	-0.369	$+0.262$	2	3/2	5/3
1	3/2	5/3	-0.054	$+0.052$	1	3/2	8/5
2	3/2	8/5	-0.054	$+0.010$	3	19/12	27/17
3	27/17	19/12	$+0.002$	-0.001	3	65/41	84/53

The leaf words are $\pi(r_0) =$ `EO` and $\pi(r_1) =$ `EEO`. Reading the word decomposition bottom-up:

Level 1 ($n = 0$, $w_1 = 2$, $\Delta > 0$, $\pi(r_3) = \pi(r_1)^2 \cdot \pi(r_0)$, $\pi(r_2) = \pi(r_1)^1 \cdot \pi(r_0)$):

$$\underbrace{\underbrace{\texttt{EEO}}_{\pi(r_1)}\,\underbrace{\texttt{EEO}}_{\pi(r_1)}\,\underbrace{\texttt{EO}}_{\pi(r_0)}}_{\pi(r_3)=\texttt{EEOEEOEO}} \qquad \underbrace{\underbrace{\texttt{EEO}}_{\pi(r_1)}\,\underbrace{\texttt{EO}}_{\pi(r_0)}}_{\pi(r_2)=\texttt{EEOEO}}$$

Level 2 ($n = 1$, $w_3 = 1$, $\Delta > 0$, $\pi(r_5) = \pi(r_3)^1 \cdot \pi(r_2)$, $\pi(r_4) = \pi(r_2)$):

$$\underbrace{\underbrace{\texttt{EEOEEOEO}}_{\pi(r_3)}\underbrace{\texttt{EEOEO}}_{\pi(r_2)}}_{\pi(r_5)=\texttt{EEOEEOEOEEOEO}} \qquad \underbrace{\underbrace{\texttt{EEOEO}}_{\pi(r_2)}}_{\pi(r_4)=\texttt{EEOEO}}$$

Level 3 ($n = 2$, $w_5 = 3$, $\Delta > 0$, $\pi(r_7) = \pi(r_5)^2 \cdot \pi(r_4)$):

$$\underbrace{\underbrace{\texttt{EEOEEOEOEEOEO}}_{\pi(r_5)}\ \underbrace{\texttt{EEOEEOEOEEOEO}}_{\pi(r_5)}\ \underbrace{\texttt{EEOEO}}_{\pi(r_4)}}_{\pi(r_7)}$$

The same estimates as mediant trees (steps $n = 1$ and $n = 2$):

$$\begin{aligned}
r_2 &= 3/2 = \text{Mediant}(\underbrace{1}_{\text{leaf}},\ \underbrace{2}_{\text{leaf}},\ w = 1),\\
r_3 &= 5/3 = \text{Mediant}(\underbrace{1}_{\text{leaf}},\ \underbrace{2}_{\text{leaf}},\ w = 2),\\
r_4 &= 3/2 = \text{Mediant}(\text{Mediant}(1,\ 2,\ 1),\ \text{Mediant}(1,\ 2,\ 2),\ w = 0),\\
r_5 &= 8/5 = \text{Mediant}(\text{Mediant}(1,\ 2,\ 1),\ \text{Mediant}(1,\ 2,\ 2),\ w = 1).
\end{aligned}$$

A weight of $w = 0$ means the left subtree is used directly (contributing zero copies of the right subtree).

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9.5 The mediant word encodes the fraction

By Lemma 9.3, the mediant word $\pi(r)$ of an estimate $r = e/o$ is uniquely determined by its fraction: $\pi(r) = \text{Ch}^-(e, o)$. The fraction e/o records the full construction history of r via the Euclidean structure of the mediant algorithm. Conversely, the fraction e/o can be read off from $\pi(r)$ by counting E's and O's. As n increases, the ratio of E's to O's in $\pi(r_{2n+1})$ converges to θ, with strict improvement guaranteed across layers (Theorem 10.1) but not within a single layer.

10 Convergence Analysis

Theorem 10.1 (Strict decrease of $|\varepsilon|$)**.** *At each iteration of the algorithm, $|\varepsilon_{i+1}|$ strictly decreases.*

Proof. By Theorem 7.1, both $|\varepsilon_j|$ and $|\varepsilon_k|$ are strictly less than $|\varepsilon_{i+1}|$. The next iterate uses the better of r_j and r_k as its r_{i+1}, so $|\varepsilon_{i+1}^{\text{new}}| < |\varepsilon_{i+1}^{\text{old}}|$. □

Remark 10.2 (Connection to continued fractions). The connection between the mediant algorithm and continued fractions is a direct consequence of the Sturmian identity (Theorem 12.6). In combinatorics on words, the standard sequence for a Sturmian word with slope $\alpha = 1/(1 + \theta)$ is generated by the recurrence $s_n = s_{n-1}^{d_n} \cdot s_{n-2}$, where the weights d_n

are precisely the partial quotients of the continued-fraction expansion of α (Lothaire [4], Theorem 2.3.5; Widmer [5], Proposition 2.4.6). The mediant word sequence agrees with this standard sequence at all interior positions (Theorem 12.6), so the optimal weights w^* track the partial quotients of α, and the denominator recurrence $o_{2n+3} = w^* \cdot o_{2n+1} + o_{2n-1}$ mirrors the classical recurrence $q_{k+1} = a_k q_k + q_{k-1}$. The algorithm thus inherits superlinear denominator growth from large partial quotients without requiring a separate proof.

However, the MT fractions are not in general the same as the CF convergents. The standard CF recurrence jumps by the full partial quotient a_k at each step, landing on a new convergent p_k/q_k. The MT algorithm picks the optimal bracketing weight w^*, which may be strictly less than a_k: in that case the new estimate is a *semi-convergent* — a Stern-Brocot fraction between two consecutive CF convergents — not a CF convergent itself. Conversely, when w^* is large it can jump past several intermediate semi-convergents in one step, skipping CF convergents that lie between them. Neither sequence is a subsequence of the other: the MT visits some semi-convergents that CF skips, and CF visits some convergents that the MT skips. The two sequences are related via the shared Sturmian word structure and produce equivalent denominator growth, but they traverse different fractions.

11 Examples and Computational Results

We illustrate the algorithm on two targets.

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11.1 Approximating $\log_2 3$

Table 1 shows the first several bounded pairs produced by the algorithm for $\theta = \log_2 3$. The initial pair is $r_0 = 1/1$, $r_1 = 2/1$ (r_1 is the better estimate throughout).

Table 1: Successive bounded pairs approximating $\log_2 3 \approx 1.58496\ldots$

n	r_{2n}	r_{2n+1}	$\varepsilon_{r_{2n}}$	$\varepsilon_{r_{2n+1}}$	w_{2n+1}	r_j	r_k
0	1/1	2/1	-0.369	$+0.262$	2	3/2	5/3
1	3/2	5/3	-0.054	$+0.052$	1	3/2	8/5
2	3/2	8/5	-0.054	$+0.009$	3	19/12	27/17
3	27/17	19/12	$+0.002$	-0.001	3	65/41	84/53
4	65/41	84/53	$+2.5 \times 10^{-4}$	-3.6×10^{-5}	6	485/306	569/359
5	485/306	569/359	$+3.0 \times 10^{-6}$	-2.7×10^{-6}	1	485/306	1054/665
6	485/306	1054/665	$+3.0 \times 10^{-6}$	-6.0×10^{-8}	24	24727/15601	25781/16266

11.2 Approximating π

For comparison, Table 2 shows the first several steps for $\theta = \pi$, starting from $r_0 = 4/1$, $r_1 = 3/1$. The classical approximant 355/113 appears at step $n = 2$ with $\varepsilon \approx +8.49 \times 10^{-8}$, after which the weight $w_5 = 293$ produces a large jump in precision in a single step.

Table 2: Successive bounded pairs approximating $\pi \approx 3.14159\ldots$

n	r_{2n}	r_{2n+1}	$\varepsilon_{r_{2n}}$	$\varepsilon_{r_{2n+1}}$	w_{2n+1}	r_j	r_k
0	4/1	3/1	$+0.273$	-0.045	7	22/7	25/8
1	25/8	22/7	-5.0×10^{-3}	$+4.0 \times 10^{-4}$	15	333/106	355/113
2	333/106	355/113	-2.7×10^{-5}	$+8.5 \times 10^{-8}$	293	103993/33102	104348/33215

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12 Relationship to Sturmian Sequences

A *Sturmian word* over $\{\mathsf{O}, \mathsf{E}\}$ with slope $\alpha \in (0,1)$ is defined by

$$s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor, \qquad n \geq 0,$$

encoding the sequence as O when $s_n = 1$ and E when $s_n = 0$ (see also Section 4). Since $\alpha < 1$, we have $s_0 = \lfloor \alpha \rfloor = 0$, so the Sturmian word always begins with E.

12.1 Structural properties of mediant words

Before stating the main Sturmian identity we record four structural results about finite mediant words.

Theorem 12.1 (Every mediant word ends in O)**.** *For every estimate r produced by the mediant tree, the mediant word $\pi(r)$ ends in O.*

Proof. By structural induction on the mediant tree.

Base case. $\pi(e/1) = \mathsf{E}^e\mathsf{O}$, which ends in O.

Inductive step. Assume $\pi(r_0)$ and $\pi(r_1)$ both end in O. If $\Delta > 0$ then $\pi(r_{\text{new}}) = \pi(r_1)^w \cdot \pi(r_0)$, whose last symbol is the last symbol of $\pi(r_0)$, which is O by hypothesis. If $\Delta < 0$ then $\pi(r_{\text{new}}) = \pi(r_0) \cdot \pi(r_1)^w$, whose last symbol is the last symbol of the final copy of $\pi(r_1)$, again O. □

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Theorem 12.2 (Δ predicts whether the last letter of $\pi(r)$ matches the Sturmian word at that position)**.** *Let $\theta > 1$ be irrational, let (r_i, r_{i+1}) be a bounded pair for θ, let $w = w^*$ be the optimal weight of Theorem 8.1 (so that* $\text{Mediant}(r_i, r_{i+1}, w^* - 1)$ *and* $\text{Mediant}(r_i, r_{i+1}, w^*)$ *bracket θ), and let $r = (e, o) = \text{Mediant}(r_i, r_{i+1}, w^*)$ (Definition 6.3). Write $\Delta = \Delta(r_i, r_{i+1})$ and $L = e + o = |\pi(r)|$. The last letter of $\pi(r)$ is always O (Theorem 12.1). Let s be the Sturmian word with slope $\alpha = 1/(1+\theta)$. The symbol s_{L-1} satisfies*

$$s_{L-1} = \begin{cases} \mathsf{O} & \text{if } \Delta > 0, \\ \mathsf{E} & \text{if } \Delta < 0. \end{cases}$$

Hence $\pi(r)[-1] = s_{L-1}$ exactly when $\Delta > 0$ (r an overestimate), and $\pi(r)[-1] \neq s_{L-1}$ exactly when $\Delta < 0$ (r an underestimate).

Proof. **Walk-state criterion.** The Sturmian word satisfies $s_j = \mathsf{O}$ iff the fractional part $f_j := \{(j+1)\alpha\}$ lies in $[0, \alpha)$: the walk on $[0,1)$ stepping by α emits O exactly when it wraps past 0. So $s_{L-1} = \mathsf{O}$ iff $f_{L-1} = \{(e+o)\alpha\} \in [0, \alpha)$.

Reduction to K. Set $K = o(e - \theta o)$ for the child $r = e/o$, so $e = \theta o + K/o$. Since $\alpha = 1/(1+\theta)$:

$$(e+o)\alpha = \frac{e+o}{1+\theta} = \frac{\theta o + K/o + o}{1+\theta} = \frac{o(\theta+1)}{1+\theta} + \frac{K/o}{1+\theta} = o + \frac{K\alpha}{o}.$$

Since o is a positive integer, $\{(e+o)\alpha\} = \{K\alpha/o\}$.

Sign of Δ characterises s_{L-1}. Write $f(w) = A + wB$ where $A = e_i - \theta o_i$ and $B = e_{i+1} - \theta o_{i+1}$. Since $w = w^*$ is the bracketing weight, $f(w^* - 1)$ and $f(w^*)$ have opposite signs, so $f(w^*)$ has the same sign as B (the sign that f tends to as $w \to \infty$). Because (r_i, r_{i+1}) is a bounded pair with r_{i+1} the better estimate, $\Delta > 0$ iff r_{i+1} is an overestimate iff $B > 0$. Hence $\operatorname{sgn}(K) = \operatorname{sgn}(e - \theta o) = \operatorname{sgn}(f(w^*)) = \operatorname{sgn}(B) = \operatorname{sgn}(\Delta)$. Let $r' = \text{Mediant}(r_i, r_{i+1}, w^* - 1)$. Since (r', r) is a bounded pair (Theorem 7.1), r is a good estimate, so $|K|/o = |e - \theta o| < 1$; with $\alpha < 1/2$, we have $|K\alpha/o| < \alpha < 1/2$, so the fractional part collapses to:

- $\Delta > 0$ (so $K > 0$): $K\alpha/o \in (0, \alpha)$, so $\{K\alpha/o\} = K\alpha/o \in (0, \alpha)$, hence $f_{L-1} \in (0, \alpha)$ and $s_{L-1} = \mathsf{O}$.
- $\Delta < 0$ (so $K < 0$): $K\alpha/o \in (-\alpha, 0)$, so $\{K\alpha/o\} = 1 + K\alpha/o \in (1 - \alpha, 1) \subset (\alpha, 1)$, hence $f_{L-1} \notin [0, \alpha)$ and $s_{L-1} = \mathsf{E}$.

Since the mediant tree produces both overestimates ($\Delta > 0$) and underestimates ($\Delta < 0$) infinitely often, both values of s_{L-1} occur infinitely often. □

Lemma 12.3 ($e > o$ for all non-leaf estimates)**.** *The leaf estimates $\lfloor\theta\rfloor/1$ and $\lceil\theta\rceil/1$ satisfy $e \geq o$ since $\theta > 1$; the lower leaf $\lfloor\theta\rfloor/1$ has $e = o = 1$ when $1 < \theta < 2$. The upper leaf always satisfies $e_{i+1} = \lceil\theta\rceil \geq 2 > 1 = o_{i+1}$ (strict). For any child $r = \text{Mediant}(r_i, r_{i+1}, w)$ with $w \geq 1$, if both parents satisfy $e_i \geq o_i$ and $e_{i+1} \geq o_{i+1}$ with $e_{i+1} > o_{i+1}$, then $e = e_i + w\, e_{i+1} > o_i + w\, o_{i+1} = o$ by positive linear combination. Hence $e > o$ holds inductively for every non-leaf estimate produced by the mediant tree.*

Theorem 12.4 (Universal EE prefix)**.** *For every non-leaf estimate r produced by the mediant tree (i.e. $|\pi(r)| > \lceil\theta\rceil + 1$), the mediant word begins $\pi(r)[0{:}2] = \mathsf{EE}$.*

Proof. By Lemma 9.3, $\pi(r) = \text{Ch}^-(e, o)$. Every non-leaf estimate $r = e/o$ lies strictly between the two leaf estimates $\lfloor\theta\rfloor/1$ and $\lceil\theta\rceil/1$, so in particular $e/o > \lfloor\theta\rfloor \geq 1$, giving $e > o$. By Definition 5.2, $\text{Ch}^-(e, o)[i] = \mathsf{O}$ iff $\lfloor(i+1)o/(e+o)\rfloor > \lfloor i \cdot o/(e+o)\rfloor$. For $i = 0$: $1 \cdot o/(e+o) < 1$ (since $o < e + o$), so $\lfloor o/(e+o)\rfloor = 0 = \lfloor 0\rfloor$, giving $\pi(r)[0] = \mathsf{E}$. For $i = 1$: $e > o$ implies $e + o > 2o$, so $2o/(e+o) < 1$, giving $\lfloor 2o/(e+o)\rfloor = 0$, and $\pi(r)[1] = \mathsf{E}$. Hence $\pi(r)[0{:}2] = \mathsf{EE}$. □

Theorem 12.5 (Palindrome invariant)**.** *For every estimate r produced by the mediant tree with $L = |\pi(r)| \geq 2$, the interior slice $\pi(r)[1{:}L{-}1]$ is a palindrome.*

Proof. By Lemma 9.3, $\pi(r) = \text{Ch}^-(e, o)$ with $L = e + o$. By Definition 5.2, $\text{Ch}^-(e, o) = \mathsf{E} \cdot U \cdot \mathsf{O}$ where U is a central word, hence a palindrome (Berstel [6], Chapter 1). Therefore $\pi(r)[1{:}L{-}1] = U$ is a palindrome. □

Theorem 12.6 (Sturmian identity)**.** *Let $\theta > 1$ be irrational, $\alpha = 1/(\theta + 1)$, and let s be the Sturmian sequence with slope α. For every estimate r produced by the mediant tree, with mediant word $\pi(r)$ of length N,*

$$\pi(r)[i] = s_i \qquad \text{for all } 0 \leq i \leq N - 2.$$

The terminal position $i = N - 1$ is excluded: by Theorem 12.1 the mediant always emits O there, while by Theorem 12.2 the Sturmian emits O or E according to the sign of Δ.

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Proof. By Lemma 9.3, $\pi(r) = \mathrm{Ch}^-(e, o)$ with $L = e + o$. By Definition 5.2, $\mathrm{Ch}^-(e, o)[i] = \mathsf{O}$ iff $\lfloor (i+1)o/L \rfloor > \lfloor i \cdot o/L \rfloor$, and the Sturmian word satisfies $s_i = \mathsf{O}$ iff $\lfloor (i+1)\alpha \rfloor > \lfloor i\alpha \rfloor$. It suffices to show $\lfloor n \cdot o/L \rfloor = \lfloor n\alpha \rfloor$ for all $1 \le n \le L - 1$.

The estimate $r = e/o$ belongs to a bounded pair (r_{2d}, r_{2d+1}) at some depth d, with the two fractions $\alpha_0 = o_0/L_0$ and $\alpha_1 = o_1/L_1$ (where $L_k = e_k + o_k$) satisfying $\alpha_0 < \alpha < \alpha_1$ and the Farey adjacency condition $o_1 L_0 - o_0 L_1 = \pm 1$. In particular o/L is one of α_0 or α_1, and $L = L_k$ for the appropriate k.

Key lemma (Farey gap). For $1 \le n \le L - 1$, no fraction with denominator $< L$ lies strictly between α and o/L. *Proof:* Suppose m/n with $0 < n < L$ lies strictly between α and o/L. Then m/n lies strictly between α_0 and α_1. By the Farey mediant property, any fraction strictly between two Farey-adjacent fractions $\alpha_0 = o_0/L_0$ and $\alpha_1 = o_1/L_1$ (with $|o_1 L_0 - o_0 L_1| = 1$) has denominator at least $L_0 + L_1 > L$, contradicting $n < L$.

Agreement. Since no fraction with denominator $< L$ lies strictly between α and o/L, for each $1 \le n \le L-1$ the two values $n\alpha$ and $n \cdot o/L$ lie in the same unit interval $(m, m+1]$ for some integer m, giving $\lfloor n\alpha \rfloor = \lfloor n \cdot o/L \rfloor$. Hence $\mathrm{Ch}^-(e, o)[i] = s_i$ for $0 \le i \le L - 2$.

Terminal position. The exclusion of $i = L-1$ follows from Theorems 12.1 and 12.2: $\pi(r)$ ends in O, while $s_{L-1} = \mathsf{O}$ iff $\Delta > 0$ (i.e. r is an overestimate). □

Corollary 12.7 (Mismatch characterisation)**.** *The mediant word $\pi(r)$ and the Sturmian prefix $s[0{:}L]$ are identical if and only if $\Delta > 0$. When $\Delta < 0$, they agree at every position except $i = L - 1$, where the mediant emits O and the Sturmian emits E.*

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The single terminal discrepancy is an artefact of truncation, not a genuine disagreement: the mediant sequence closes its last primitive word with O at position $N - 1$, whereas the Sturmian word — which continues indefinitely — may still be in the interior of a longer word at that depth. As the depth increases and $N \to \infty$, both sequences converge to the same infinite word.

This has been verified computationally for several irrational values of θ, including $\sqrt{2}$, $\log_2 3$, $\log_2 5$, ϕ, and π. For $\theta = \log_2 3$ the verification extends to sequences of length up to 27,869,189 symbols: at every depth the two sequences agree everywhere except possibly the last symbol, where the mediant emits O and the Sturmian emits E.

Remark 12.8 (The mediant sequence is the better finite approximation). At depths where the two sequences differ, the mediant's finite mediant word is a closer approximation to $\alpha = 1/(\theta + 1)$ in terms of running O-density. The terminal O emitted by the mediant sequence gives a density slightly *above* α, whereas the Sturmian's terminal E gives a density substantially *below* α.

This reflects a structural asymmetry in the two methods. The Sturmian algorithm at position $N - 1$ is mid-word in a longer primitive word, so it emits E — inherently favouring the above-θ estimate and leaving the running density below α. The mediant tree, by contrast, explicitly selects at each step the estimate with smaller $|\varepsilon|$, which is the lower-bound estimate r_{2n} (the below-θ member of the pair) when that is the better of the two. Closing the sequence with O at that point reflects this selection: it is the correct terminal for a word whose ratio e/o slightly underestimates θ, nudging the running density up toward α.

For $\theta = \log_2 3$, the improvement at the discrepancy depths is substantial: at $N = 1719$ the mediant density error is 1.4×10^{-8} against the Sturmian's 5.8×10^{-4} — roughly four orders of magnitude better.

The O-density of $\pi(r)$ for an estimate e/o is

$$\frac{o}{e+o} = \frac{1}{e/o+1} \xrightarrow{e/o\to\theta} \frac{1}{\theta+1} = \alpha,$$

consistent with the Sturmian slope.

Efficiency and lazy generation. Although the mediant words are Sturmian, the mediant tree is a far more efficient way to generate them. To produce a prefix of N symbols, the naive Sturmian algorithm must evaluate $s_n = \lfloor (n+1)\alpha \rfloor - \lfloor n\alpha \rfloor$ at each of the N positions, requiring $O(N)$ real-arithmetic operations.

The mediant algorithm instead builds a compressed representation of the same prefix. Each iteration requires a constant number of real-arithmetic operations to compute the next bounded pair and weight; after k iterations the tree encodes a Sturmian prefix of length $N = e_{2k+1} + o_{2k+1}$, which grows exponentially in k. Thus only $O(\log N)$ real-arithmetic operations are needed to encode a prefix of length N.

Once the tree is built, enumerating the N symbols requires *no real arithmetic at all*: each node specifies an integer weight w and two child words, and reading off the sequence is a recursive depth-first traversal governed entirely by integer counts. No auxiliary memory beyond the call stack is required.

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OO-avoidance. Since $\theta > 1$ implies $\alpha = 1/(\theta+1) < 1/2$, the Sturmian word avoids OO: two consecutive O's would require $\lfloor (n+2)\alpha \rfloor - \lfloor n\alpha \rfloor = 2$, but $2\alpha < 1$ makes this impossible. The mediant tree gives an independent structural proof: every mediant word ends in O and begins with E (since $e > o$ for all non-leaf estimates; Lemma 12.3), so every concatenation boundary within a branch word $v^w \cdot u$ or $u \cdot v^w$ reads $\cdots\mathsf{OE}\cdots$; OO never arises at any boundary, and by induction never arises anywhere.

Remark 12.9. The weights w_{2n+1} are related to the partial quotients of the continued fraction expansion of θ, though they are not in general identical to them. A large weight at step n corresponds to a large partial quotient, producing a long run of one symbol type in the mediant word — the same phenomenon that gives rise to the irregular spacing of O's and E's in an irrational Sturmian word. Conversely, small weights correspond to small partial quotients, and the mediant word advances by only a short block at each such step.

13 Extension to $0 < \theta < 1$

The algorithm of Sections 7–9 requires $\theta > 1$. We now show how to handle $\theta \in (0, 1)$ by algebraic reduction, in two stages.

If one were content to accept OO in the mediant word, no special treatment would be needed: the algorithm and the Sturmian definition extend formally to any $\theta > 0$, and for $\theta < 1$ the slope $\alpha = 1/(1+\theta) > 1/2$ makes OO unavoidable. The reductions below are

motivated by applications (described in Appendix **??**) that require OO-free mediant words. More concretely: for $\theta \in (0,1)$, setting $\theta' = (1-\theta)/\theta$ (when $\theta < 1/2$) or $\theta' = \theta/(1-\theta)$ (when $\theta > 1/2$) maps θ to $\theta' > 1$, giving a slope $1/(1+\theta') < 1/2$ and hence an OO-free sequence; the counts e and o from that sequence then yield a rational approximant to the original θ.

13.1 Case $0 < \theta < \frac{1}{2}$

For $\theta \in (0, \frac{1}{2})$, define

$$\theta' = \frac{1-\theta}{\theta} = \frac{1}{\theta} - 1 > 1.$$

Run the algorithm on θ'. A bounded pair $e/o \approx \theta'$ satisfies $e/o \approx (1-\theta)/\theta$, so

$$\theta \approx \frac{o}{e+o}.$$

The approximant to θ is therefore $o/(e+o)$, with numerator o (the O-count of the mediant word) and denominator $e+o$ (its total length).

Mediant word. The mediant word for θ' has O-density

$$\frac{1}{1+\theta'} = \frac{1}{1+(1-\theta)/\theta} = \theta,$$

which is less than $\frac{1}{2}$ since $\theta < \frac{1}{2}$. Consequently the OO-avoidance result of Section 12 applies: the mediant word for θ' is OO-free, and its O-density converges to θ.

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Sturmian identity. The Sturmian word with slope $\alpha' = \theta$ is the infinite mediant word produced by the mediant tree for θ' (Theorem 12.6). This Sturmian word encodes θ directly: the O-density of any long prefix approximates θ with the same precision as the underlying mediant approximation of θ'.

13.2 Case $\frac{1}{2} < \theta < 1$

For $\theta \in (\frac{1}{2}, 1)$, note that $1-\theta \in (0, \frac{1}{2})$, so define

$$\theta' = \frac{\theta}{1-\theta} > 1$$

and run the algorithm on θ'. A bounded pair $e/o \approx \theta'$ satisfies $e/o \approx \theta/(1-\theta)$, so

$$\theta \approx \frac{e}{e+o}.$$

No modification of the mediant word is required: the sequence for θ' is used solely to produce the counts e and o, which then yield the approximant $e/(e+o)$ to the original θ.

Mediant word. The mediant word for θ' has O-density

$$\frac{1}{1+\theta'} = \frac{1}{1+\theta/(1-\theta)} = 1-\theta < \tfrac{1}{2},$$

so it is OO-free and its Sturmian slope is $1-\theta$. The sequence encodes θ'; the rational approximant $e/(e+o)$ to θ is read off from the E- and O-counts of that sequence.

Summary. Table 3 collects the reductions for all four ranges of θ.

Table 3: Reductions to the $\theta > 1$ algorithm for all ranges of θ

Range of θ	Reduction θ'	Approximant to θ	O-density
$\theta > 1$	$\theta' = \theta$	e/o	$1/(1+\theta) < \frac{1}{2}$
$0 < \theta < \frac{1}{2}$	$\theta' = (1-\theta)/\theta$	$o/(e+o)$	$\theta < \frac{1}{2}$
$\frac{1}{2} < \theta < 1$	$\theta' = \theta/(1-\theta)$	$e/(e+o)$	$1-\theta < \frac{1}{2}$
$\theta < 0$	$\theta' = -\theta$ (or above)	negate	—
$\theta = 0$	trivial: word E, $o/(o+e) = 0$	$0/1$	—
$\theta = \frac{1}{2}$	trivial: word EO, $o/(o+e) = \frac{1}{2}$	$1/2$	—
$\theta = 1$	trivial: word O, $o/(o+e) = 1$	$1/1$	—

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14 Conclusion and Open Problems

We have presented a theory of rational approximation for real numbers based on bounded pairs and the weighted mediant construction. The core algorithm applies to $\theta > 1$; all other real numbers — including the boundary cases $\theta \in \{0, \frac{1}{2}, 1\}$, whose mediant words are the trivial words E, EO, and O respectively — are handled by algebraic reduction or direct construction (Section 13). The central result, Theorem 8.1, gives a closed-form formula for the optimal mediant weight at each step, enabling an efficient iterative algorithm whose denominator sequence grows in proportion to the optimal weights — superlinearly when partial quotients are large. The connection to continued fractions falls out as a consequence of the Sturmian identity: the optimal weights track the partial quotients of $1/(1+\theta)$, giving superlinear denominator growth at no extra proof cost (Remark 10.2). The mediant word (Section 9) is uniquely determined by its fraction, and the sequence converges to the Sturmian word with slope $1/(\theta+1)$, agreeing at all interior positions of every finite prefix, with the terminal symbol fully characterised by the sign of Δ (Theorem 12.6). Two structural invariants further characterise mediant words: every non-leaf word begins with EE (Theorem 12.4), and the interior slice $\pi(r)[1{:}L{-}1]$ is a palindrome for any mediant word of length $L \geq 2$ (Theorem 12.5). Section 13 extends the construction to all $\theta \in (0,1)$ via the reductions $\theta' = (1-\theta)/\theta$ (for $\theta < \frac{1}{2}$) and $\theta' = \theta/(1-\theta)$ (for $\theta > \frac{1}{2}$); in both cases the algorithm runs on the reduced $\theta' > 1$ and the approximant to θ is read off from the resulting e and o counts.

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A Proof of Theorem 8.1

We prove the closed-form formula $w^* = \lfloor -A/B \rfloor + 1$ directly from the bracketing condition, using only a linear sign argument.

We have a bounded pair (r_i, r_{i+1}) bracketing θ, with

$$A = e_i - \theta\, o_i, \qquad B = e_{i+1} - \theta\, o_{i+1}.$$

Since r_i and r_{i+1} are on opposite sides of θ we have $\operatorname{sgn}(A) = -\operatorname{sgn}(B)$, so $A/B < 0$ and $c := -A/B > 0$. The two weighted mediants are

$$r_j = \frac{e_i + (w-1)\, e_{i+1}}{o_i + (w-1)\, o_{i+1}}, \qquad r_k = \frac{e_i + w\, e_{i+1}}{o_i + w\, o_{i+1}},$$

with signed errors (numerators of ε_j, ε_k up to positive denominators)

$$A + (w-1)B \qquad \text{and} \qquad A + wB.$$

Bracketing condition. The pair (r_j, r_k) brackets θ iff ε_j and ε_k have opposite signs, i.e. $(A+(w-1)B)(A+wB) < 0$. Writing $c = -A/B$, this product equals $B^2\,(w-1-c)(w-c)$, which is negative iff $w-1-c$ and $w-c$ have opposite signs, i.e.

$$w - 1 < c < w \qquad \Longleftrightarrow \qquad c - 1 < w - 1 \le c \qquad \Longleftrightarrow \qquad w = \lfloor c \rfloor + 1.$$

Since θ is irrational, $A + nB \neq 0$ for all integers n, so c is never an integer and the condition is strict: exactly one integer w satisfies it, namely

$$w^* = \lfloor -A/B \rfloor + 1.$$

Expression in terms of Δ and K. Recall $K = o_{i+1}B$ and $A\,o_{i+1} - B\,o_i = -\Delta$, giving $A = (Bo_i - \Delta)/o_{i+1}$, so

$$\frac{-A}{B} = \frac{\Delta}{o_{i+1}B} - \frac{o_i}{o_{i+1}} = \frac{\Delta}{K} - \frac{o_i}{o_{i+1}},$$

and therefore

$$w^* = \left\lfloor \frac{\Delta}{K} - \frac{o_i}{o_{i+1}} \right\rfloor + 1,$$

completing the proof of Theorem 8.1.

Termination. When $K = 0$, r_{i+1} is an exact rational representation of θ; no further improvement is possible and the algorithm terminates.

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